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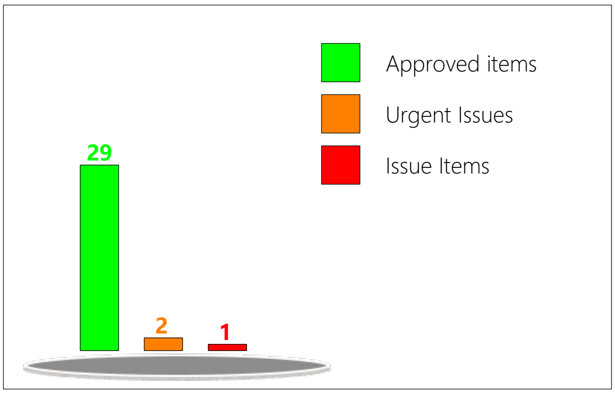
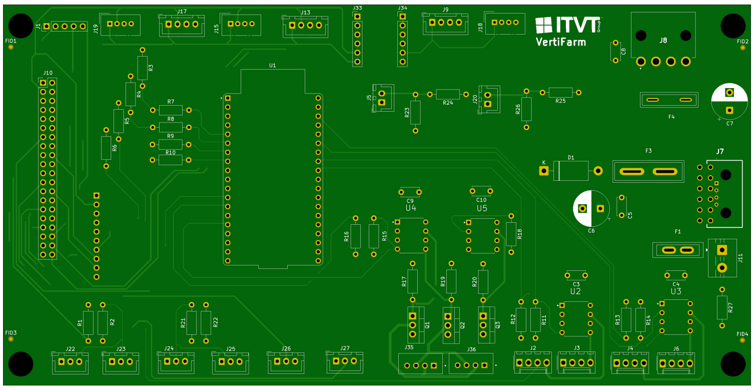
HQDFM Design for Manufacture(DFM) Report

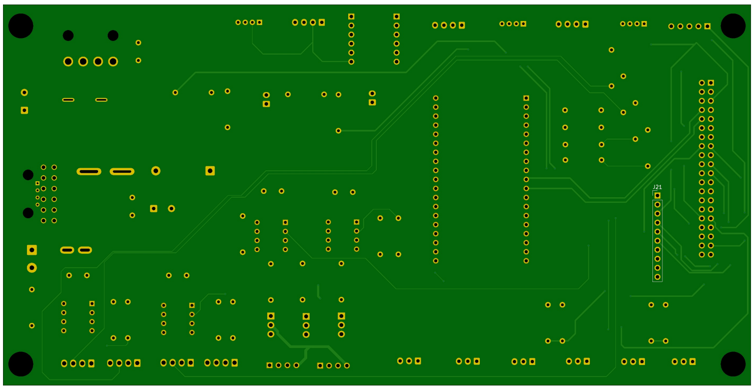
File name: VertiFarm_PCB

Time: 2026-08-20Layer count:4 PCB Thickness: 1.60 Quantity: 5 mm



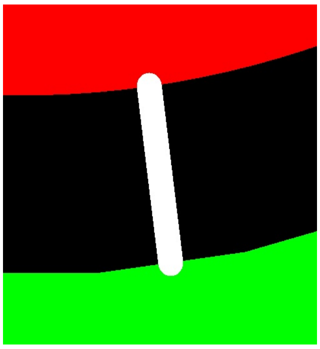
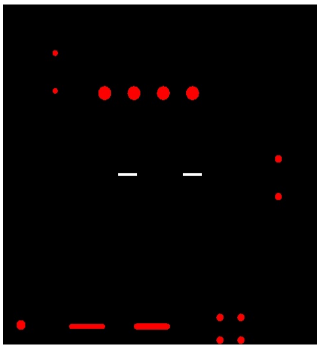
Basic Board Specs	Trace Width/Spacing	8.00/8.00mil
	Milling Density	31.5003m/m²
	Surface Finish Area	11.39%
	Test Point Count	328





Type	Category	No. of Checks	Result
PCB Trace Analysis	Open/Shorts (IPC)	1	Fail
	Signal Integrity	4	Pass
	Smallest Trace Width	1	Pass 3
	Smallest Trace Spacing	3	Pass 14
	SMD Pad Spacing	1	Pass
	Pad Size	3	Pass 2
	Hatched Copper Pour	2	Pass
	Annular Ring Size	2	Pass 6
	Drill to Copper	5	Pass 43 , Fail 3
	Copper-to-Board Edge	2	Pass 20
	Holes on SMD Pads	4	Pass
PCB Drilling Analysis	Drill Diameter	8	Pass 54 , Fail 1
	Drill Spacing	4	Pass
	Drill to Board Edge	4	Pass
	Drill Hole Density	1	Pass
	Special Drill Holes	2	Pass
	Drill Hole Errors	3	Pass
PCB Solder Mask Analysis	Solder Mask Dam	2	Pass
	Missing SMask Opening	1	Pass
	Solder Paste Area	1	Pass
PCB Silk Analysis	Silkscreen Spacing	1	Pass 3

PCBA Component Analysis	Component Spacing	1	Fail
	Comp.-to-Board-Edge	3	Fail
	Component Silkscreen Spacing	0	Fail
	Pad Count Mismatch	4	Fail
	Designator Length	0	Fail
	Double-sided Components	1	Pass
	Mid Mount Comp	1	Pass
	Comp Height Analysis	2	Pass
PCBA Pin Analysis	SMT Pad Analysis	8	Fail
	Through-hole Pins	10	Fail
PCBA Pad Analysis	Chip Pads	60	Fail
PCBA Fiducial Analysis	Fiducial Count	1	Fail
	Fiducial Analysis	4	Pass

ID	Check	Limits	Value	Issue	Image	Position	Qty	Level
1	Drill to Copper_NPTH-to-Copper	8,10,12	0.25 mm	NPTH to copper spacing of 9.86mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		235.16,-95.46	3	Warning
2	Drill Diameter_Smallest Slot Width	0.45,0.6,0.7	0.30 mm	Plated Minimum Slot Width:0.46mm were detected in your design. Smaller drilling bits are more likely to break and need to be replaced frequently, which decrease manufacturing efficiency and and take longer to produce.They should be enlarged to at least 0.6mm.		219.03,-72.77	4	Warning